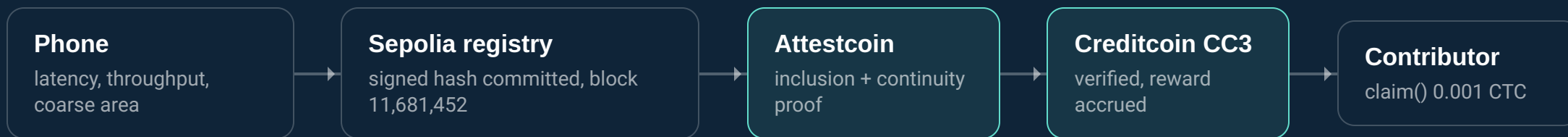




SIGNALPROOF

Connectivity data a contract can check.

Phones measure real network conditions. Each measurement is committed on Ethereum Sepolia, proven by the Attestcoin Protocol, and settled with a reward on Creditcoin.



BUIDL CTC 2026 Fall

DePIN track

Deployed and verified on Sepolia and CC3 Testnet

Anyone can claim coverage numbers. Nobody can prove them.

Operators publish coverage maps. DePIN networks pay contributors for connectivity data. Venues and public programs buy both. None of them can audit what they are paying for.

The data is self-reported. A coverage claim is a number someone typed. The device that produced it, the place, the time — none of it is checkable after the fact.

The incentive is to inflate. Where contributors are paid per sample, a fake sample is worth exactly as much as a real one until someone can tell them apart.

Location data is the price of honesty today. The usual fix is to collect precise GPS trails, which no one should have to hand over to prove their phone had two bars.

What SignalProof changes

The measurement becomes an event on a public chain, the event is proven to a second chain by the Attestcoin Protocol, and a contract there decides — from the proof alone — whether it is real enough to pay for.

Not the gateway. Not the relayer. Not the person who ran the test. The proof.

One measurement, two chains, one decision.

Everything before the Sepolia event decides what is **admitted**. Everything after it decides what is **paid**. The two never trust each other.

Measure

Browser client:
median of 7 timed
pings, a 3 MB
download, the
network class, and a
geohash-6 area. No
coordinate leaves the
device.

seconds

Admit

Gateway recomputes
the measurement root,
recovers the
contributor's signature,
checks freshness and
uniqueness, then the
relayer commits the
hash to
SourceBatchRegistry.

next 15 s tick

Attest

Creditcoin's
attestation of Sepolia
catches up to the
block. The worker
polls the attested
height once per tick
instead of blocking.

7–9 min

Prove

ProofBuilder returns
the Merkle inclusion
proof and the
continuity proof for
the transaction.

7 siblings, 6
roots

Settle

SignalProofSettlement
verifies through the
BlockProver precompile,
applies its own checks,
and accrues 0.001 CTC to
the contributor.

150,640 gas

Recorded end to end: 8.5 minutes, from the Sepolia transaction to a claimable reward. The attestation wait is the whole cost; everything else is seconds.

Payment is permissionless. Anyone holding a valid proof may settle it. All authorisation lives in the contract's checks, so a relayer that disappears cannot strand a measurement.

What the precompile proves, and the two things it does not.

SignalProofSettlement inherits ASCBase and overrides one hook. The BlockProver precompile proves inclusion in an attested Sepolia block — not that the transaction succeeded, and not which contract emitted its logs.

Protocol detail	What we do
Receipt status	Reject unless receiptStatus == 1. Inclusion is not success.
Emitter binding	Reject a log whose emitter is not our registry. Missing from the docs' SimpleMinterASC example, present in Gluwa's ASCLoanManager; mirrored here with a whole-system test where a forged registry holds a valid proof.
Contributor signature	Recovered on-chain by the registry from the exact EIP-191 text the wallet signed. A relayer that names another payee holds no signature for that pairing.
chainKey	Resolved at startup from getSupportedChains() on chainId. It is a uint64; chainName is hex-encoded; key 1 means another chain on Mainnet.
Attestation wait	Poll the attested height once per tick; the SDK's blocking wait would freeze the queue for 8–20 minutes.

Two replay layers

ASCBase deduplicates by queryId; the contract also refuses a measurementRoot it has already settled.

Batch route

Many measurements, one continuity proof, one transaction. A bad entry is skipped and reported, never fatal to the honest ones beside it.

Pull-payment rewards

No value moves on the verification path, so a reverting receiver cannot block settlement for anyone else.

Deployed, verified, and settled on-chain.

Contract	Chain	Address
SourceBatchRegistry (signature on-chain)	Sepolia 11155111	0x32c0923cD58523864D2727FCaaB109783664c236
SignalProofSettlement	CC3 Testnet 102031	0x8F14B2cC1b807203d332DE6E3DA6274176FDb584
SignalProofBatchSettlement	CC3 Testnet 102031	0x3B90e22f246bBa68f6de682b564c33b121D68C85

First run through the signed registry	Result
Signed measurement committed to Sepolia	0xbd7261...7223 block 11,681,452
Attestation reached that block	9.6 min, advancing 10 blocks at a time
Inclusion proof	7 Merkle siblings, 9 continuity roots, 2,112-byte txBytes
Settled on Creditcoin	0x60d9e4...a2b5 block 5,468,910, 154,224 gas
Earlier: claim()	0x148536...e0030 pool 5.000 to 4.999 CTC

9.8 minutes end to end

8.5 on the earlier run; the attestation wait is the whole cost

3 in 1 batch settlement

Three measurements, one continuity proof, one CC3 transaction — 0xcf1a01...c49598 and 0xb0933e...32d990.

75 Foundry tests

127 server tests

plus a live smoke test of the Attestcoin read path that passes from a clean clone.

Evidence — all transactions are linked in the README

SignalProof

Four ways to get paid for nothing, found and closed before submission.

Each exploit is kept in the test suite with its assertions inverted, so reopening the hole fails a test instead of quietly working.

1

The registry was permissionless

Anyone could call the real registry, name themselves as payee, obtain a genuine proof and settle it. Emitter binding passes, because the event really came from the bound registry. Fixed by gating admission to the relay that carries the gateway's checks. Redeployed.

`AdvUnlimitedMint.t.sol`

2

One measurement paid twice

Two settlement routes, one registry, two `settled` maps. The same genuine proof could be paid from both pools. The batch contract now defers to any root a sibling already paid, via a guarded `staticcall` that degrades to no cross-check rather than refusing everyone.

`DoubleSettlement.t.sol`

3

One poisoned log killed a batch

A Sepolia transaction that touched the registry and emitted a lookalike log made the batch revert forever, taking the honest measurements down with it. Foreign logs are now skipped and reported through `ForeignLogSkipped`. `BatchPoison.t.sol`

4

The relayer could name the payee

Gating on the relayer left one party fully trusted: it could commit a genuine event naming anyone as contributor. The registry now recovers the contributor's EIP-191 signature on-chain; a vector signed with ethers is accepted by the contract.

`SourceBatchRegistry.t.sol`

Why this matters for Attestcoin depth. The precompile was never wrong in any of these; every exploit used a genuine proof. Admission, attribution, replay and batching are where cross-chain security is won or lost — that is where this project spent its time.

Batching is cheaper per entry, and it gets more expensive as it grows.

Two live runs are not enough to state a saving, so `BatchGasProbe.t.sol` holds the proof shape fixed and varies only the batch size.

Entries	Total gas	Marginal gas per extra entry
1	77,828	—
10	482,141	44,892
50	2,468,789	51,015
200	12,476,653	71,165

Measured, not estimated

A marginal entry costs 45–71k gas against 78k for a batch of one — and the number rises with size, because the decoder allocates memory it never reclaims and the quadratic memory-expansion term takes over.

So the worker caps a batch at 50, not at the block ceiling. The probe puts that ceiling at 600 entries for a 75,000,000-gas CC3 block; 800 does not fit.

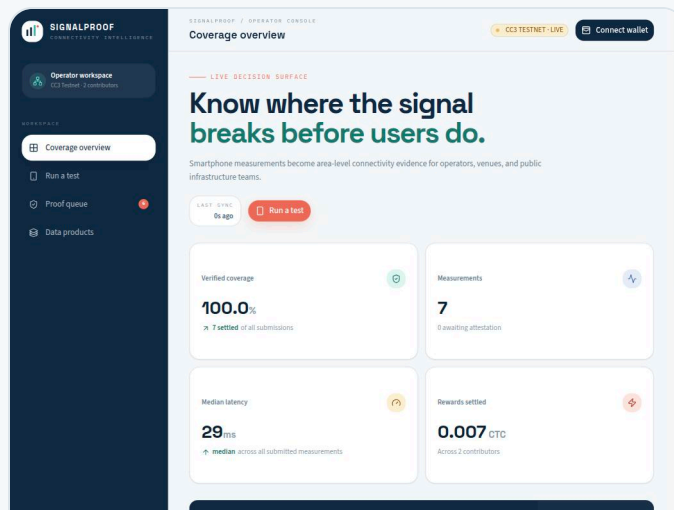
Live numbers agree in shape: 253,284 and 423,360 gas for three measurements, against 150,640 for one. The spread comes from the proof the attestation service happens to return, not from the batch.

SignalProof

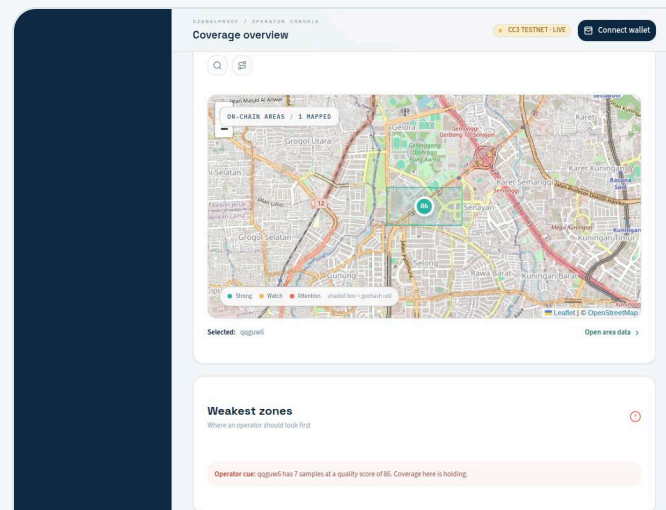
The dashboard reads the chains, not a database.

Every number below is a join of `MeasurementSubmitted` on Sepolia with `MeasurementVerified` ON Creditcoin. A keyless clone shows the same thing, and a buyer can query it:

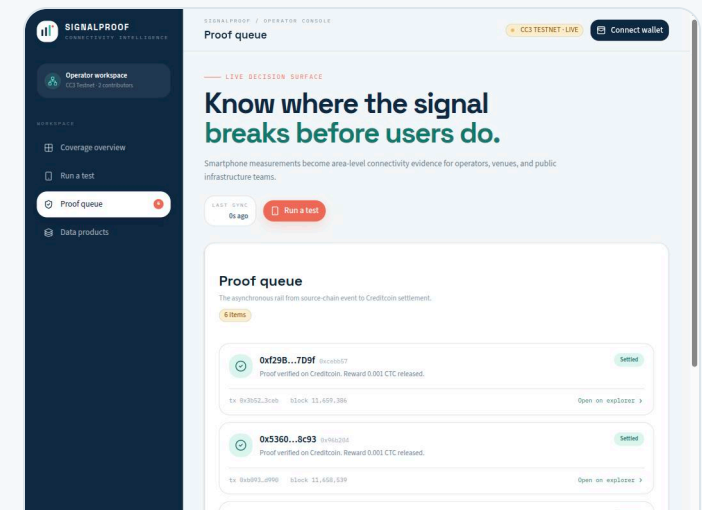
`/v1/areas/qgguw6` returns the aggregates and every sample with its two transactions.



Coverage overview. Verified share, settlements, rewards, and the four-stage proof spine, all live.



Coverage map. The shaded box is the geohash cell itself, 1.2 km by 0.6 km. Undecodable areas are listed as unmapped, never placed.



Proof queue. Each row opens the live Attestcoin proof — attested height, Merkle path, continuity roots, the exact `execute()` call — and can be settled from the contributor's own wallet.

Buyers pay contributors. Anyone can check the receipt.

A buyer sends 0.05 CTC to `SignalProofSettlement` — its `receive()` credits the pool that contributors `claim()` from — signs the transaction hash, and receives a stateless key. First purchase on chain: pool 4.997 → 5.047 CTC. Metered is the service; the data is public on two chains, and the dashboard, catalog, verifier and badges stay free.

API ACCESS

0.05 CTC for 30 days, paid into the reward pool.
The payment is a plain CTC transfer to 0x8F14B2...b584 — the settlement contract's own balance, which contributors `claim()` from. You are paying the people who measured. The key unlocks per-area samples with provenance, briefs and exports; the dashboard and the catalog stay free, because the data itself is public on two chains. What is metered is the service.
1. Your wallet sends 0.05 CTC on Creditcoin CC3 Testnet (gas is yours).
2. You sign one message naming the transaction — proof that the payer is you.
3. The server verifies the payment on chain and returns a key derived from it. Same payment, same key: sign again to recover it.

Key active until 10/12/2026

sp1_1791804870_0xc701f96e61d9eba2dcda184915746012de08225f3e34a2ac8db0a814f997fef_2690413e3820fcdaf03f05765713877

Copy key Payment on Blockscout Forget on this browser

Buy API access. Pay → sign → key, all from the buyer's wallet. The same payment always returns the same key: nothing is stored, no replay to defend.

Surface	What it does	Cost
/verify/<hash>	root, Sepolia tx or Creditcoin tx → both chains and the Attestcoin proof	free
/area/<geohash>	cell on the map, quality over time, every sample; CSV · JSON · brief · badge	key · badge free
/contributors · /ops	leaderboard and per-address history; relayer gas, pool runway, worker and RPC health. Auto-measure: every 10 min, each cycle wallet-signed	free

MATCHED BY CREDITCOINTXHASH

SETTLED

Settled on Creditcoin
Area `qqguw6` · contributor 0x536029E7...e68c93 · 9/11/2026, 6:52:33 PM

Measurement root

What the contributor signed. Latency, throughput, area, timestamp and a nonce hash to this value.
0xf346bef13717f9df46fd75e8c183ff1517cc2575863a71de4b8209b855712a0

Committed on Ethereum Sepolia

Block 11,681,452 · MeasurementSubmitted emitted by the registry, which recovered the contributor's signature on-chain.
0xbd7261e655c751f9a38acc116808acd3a0c0b0fd14279fe4fad207c742d57223 explorer

Attested by Creditcoin

Creditcoin's attestation frontier passed this block; the proof below was built from it.

Settled on Creditcoin CC3

MeasurementVerified · reward 0.001 CTC accrued to the contributor after the precompile verified inclusion and the contract checked receipt status, emitter and signature.
0x60d9e438e8549a101b144bd1e474475301c696640062601ddfd18f788e6aa2b5 explorer

Attestcoin proof — what the precompile verified

chainKey 1 · verifier 0x0000_0F02

ATTESTED HEIGHT	11,681,452 (Sepolia block, tx index 96)
TRANSACTION	2,112 bytes of RLP-encoded transaction + receipt — the precompile decodes the receipt status and the MeasurementSubmitted log from these
MERKLE INCLUSION	root 0x145d2dde... · 7 siblings: R 0x8b7743b1..., R 0xbf56408f..., R 0x9c7ed819... +4 more
CONTINUITY	49 roots from lower endpoint 0x3e600904... — the walk from this block back to an attestation checkpoint

Public verifier. Commitment, attestation, proof, settlement — each with its transaction. Never metered.

Screenshots taken on 2026-09-13 from the live testnet deployment · pnpm e2e: surface 28/28

SignalProof

Coarse by construction. Measured, not asked.

Location never leaves the phone

The area is a geohash of at most six characters, enforced by the gateway, stored on-chain as `bytes32`. A cell is roughly **1.2 km by 0.6 km**; a contributor is somewhere inside the box, never at a dot. The raw coordinate is converted in the same function that reads it and is never transmitted.

The wallet is an identity, not a signer

The relayer pays gas on both chains. The connected address only decides who accrues the reward. The one transaction a contributor ever sends is `claim()`.

Why we measure instead of asking the browser

	Measured by us	Reported by <code>navigator.connection</code>
Throughput	40.8 Mbps	7.5 — capped at 10 by Chromium
Latency	68 ms median, 7 samples	50 — rounded to 50 ms
Carrier	not collected	no API exposes it
5G	not claimed	no such value exists

The UI used to show “5G · Telkomsel”. Both halves were fabricated, so both were deleted. What no browser can provide, the product does not display.

What is not done, stated before a judge finds it.

Known limitations

- The relay is still the admission point: if it goes offline, new measurements wait. It can no longer forge attribution — the registry checks the contributor's signature on-chain.
- Read direction only, Sepolia to Creditcoin. Write-ability has no public reference implementation and has not cleared third-party audit.
- Attested is not finalized: attestation runs 65–85 blocks ahead of Sepolia's finalized tag. A deep reorg could in principle invalidate an attested height.
- End to end is 9–13 minutes. A live real-time demo is not possible; the demo pre-warms a proof.
- Anti-Sybil is a rate limit (3 per contributor per cell per 10 min), not a defence; no platform attestation; the browser client is the measurement client, and auto-measure needs a wallet signature per cycle by design.
- Buyer keys are stateless: no per-key revocation (rotating the secret revokes all), and no retention policy yet.

Next 90 days

- Buyer API v2: per-key revocation, a retention policy, and cell-level subscriptions and webhooks on top of the metered v1 that ships today.
- Native measurement client with background sampling and device attestation.
- Sybil resistance beyond the rate limit: device attestation, stake-weighted rewards, cell-level anomaly detection.
- Write-ability once an audited reference implementation exists, so settlement can push results back to the source chain.

Why this belongs on Creditcoin.

The buyer is real. Operators planning capacity, venues negotiating with carriers, and public infrastructure programs auditing coverage commitments all pay for connectivity data today, and none of it is verifiable.

The proof is the product. A coverage snapshot whose every sample can be traced to an attested Sepolia event and a Creditcoin settlement is a different good from a spreadsheet of claims.

Creditcoin is where it settles. Rewards accrue in CTC, the settlement contract is the source of truth, and the Attestcoin Protocol is the reason no oracle operator sits in the middle.

Links

Repository	github.com/mdlog/signalproof
Demo video	linked from the BUIDL page on DoraHacks
First paid access	creditcoin-testnet.blockscout.com/tx/0xed00...8d30
Settlement contract	creditcoin-testnet.blockscout.com/address/0x8F14...b584
Source registry	sepolia.etherscan.io/address/0x32c0...c236
Protocol	docs.attestcoin.org

Built on the Attestcoin Protocol by Gluwa. Contract patterns follow [gluwa/attestcoin-protocol-examples](#) (Apache-2.0). MIT licensed.